Channel Estimation for Rydberg Atomic Receivers

September 1, 2026

1 Purpose and Scope

This document is a technical reference for the paper *"Channel Estimation for Rydberg Atomic Receivers"* by Xu *et al.* It follows the paper's progression: atomic receiver model → nonlinear magnitude measurement → strong-reference approximation → gradient-based channel estimation → 2D low-rank model → projected gradient descent (PGD) → CRLB → simulation figures.

The paper contains a key conceptual transition:

Nonlinear magnitude-only atomic measurement
⇓
Strong-reference linearized real-valued model
⇓
Optimization-based channel estimation

The reference paper used for understanding the GS baseline is *"Towards Atomic MIMO Receivers"*. Its biased GS algorithm is discussed separately because the channel-estimation paper cites it as the conventional phase-retrieval baseline.

1.1 Central research problem

In a conventional pilot-based MIMO receiver one often observes a linear model such as

$$Y = GS + N.$$

Here, the Rydberg atomic receiver fundamentally measures a magnitude. Therefore the corresponding raw observation is nonlinear:

$$y_{i,p} = \left| \sum_{k=1}^{K} g_{i,k} s_{k,p} + b_{i,p} + n_{i,p} \right|.$$

The unknown complex channel is hidden inside an absolute-value operation. Hence ordinary least-squares channel estimation cannot be directly applied.

2 Notation and Dimensions

Symbol	Dimension	Meaning
K	scalar	Number of single-antenna users.
I	scalar	Number of vapor-cell elements in a 1D atomic array.

I_1, I_2	scalar	Number of vapor cells along the two dimensions of a 2D array.
P	scalar	Number of pilot time slots.
L_k	scalar	Number of propagation paths for user k.
$g_{i,k}$	complex scalar	Channel from user k to 1D vapor-cell element i.
$g_{i_1,i_2,k}$	complex scalar	Channel from user k to 2D element (i_1, i_2).
G	$I \times K$	1D channel matrix.
$\mathcal{G}$	$I_1 \times I_2 \times K$	2D channel tensor.
$G_{(3)}$	$K \times I_1 I_2$	Mode-3 unfolding of $\mathcal{G}$.
S	$K \times P$	Known pilot matrix.
b or B	complex	Known reference signal/source contribution.
Z	complex	Unit-magnitude phase compensation matrix, $e^{-j\angle b}$.
Y	real	Atomic magnitude measurements after approximation.
N	real/complex	Noise, depending on the stage of the model.
μ_{eg}	vector	Atomic transition dipole moment.
$\boldsymbol{\epsilon}$	vector	Polarization direction.
$\alpha_{l,k}$	complex scalar	Gain of path l for user k.
ϕ, θ	scalar	Azimuth/elevation angles as appropriate.
$\circ$	operator	Hadamard (elementwise) product.
$\otimes$	operator	Kronecker product.
$\times_3$	operator	Tensor multiplication along mode 3.
$\mathrm{mat}(\cdot)$	operator	Reshape a vectorized row into an $I_1 \times I_2$ matrix.
$\mathrm{vec}(\cdot)$	operator	Vectorization.
$\|\cdot\|_F$	scalar	Frobenius norm.

3 Atomic Receiver Preliminaries

3.1 Equation (1): two-level atomic state

The state of an atom in vapor cell i is

$$|\Phi(t)\rangle_i = \zeta_{g,i}(t)\,|g\rangle + \zeta_{e,i}(t)\,|e\rangle\,, \tag{1}$$

with normalization

$$|\zeta_{g,i}(t)|^2 + |\zeta_{e,i}(t)|^2 = 1. \tag{2}$$

Meaning: $\zeta_{g,i}$ and $\zeta_{e,i}$ are probability amplitudes. Their squared magnitudes give the probabilities of observing the atom in the ground and excited states.

3.2 Equation (2): Schrödinger equation

$$i\hbar \frac{\partial\,|\Phi(t)\rangle_i}{\partial t} = \left(\hat{H}_i + \hat{V}_i\right)|\Phi(t)\rangle_i\,. \tag{3}$$

Here $\hat{H}_i$ is the free Hamiltonian and $\hat{V}_i$ describes interaction with the incident electromagnetic field.

Interpretation. This is the microscopic origin of the communication model: the incident RF field changes the atomic quantum state. The photodetector eventually converts the atomic response into a measurable quantity.

Important modeling transition. The remainder of the channel-estimation paper does not numerically solve (3) during every optimization iteration. Instead, it moves to an effective communication-level channel model.

4 1D Atomic Array Channel and Measurement Model

4.1 Equation (7): multipath channel

The paper models the channel between vapor cell i and user k as

$$g_{i,k} = \sum_{l=1}^{L_k} \frac{1}{\hbar} \mu_{eg}^T \epsilon_{i,k,l} \alpha_{l,k} e^{-j(i-1)\phi_{l,k}}. \tag{7}$$

Meaning of each term.

- $g_{i,k}$: effective channel coefficient between user k and vapor cell i.
- L_k: number of propagation paths associated with user k.
- $\hbar$: reduced Planck constant.
- μ_{eg}: atomic transition dipole moment vector.
- $\epsilon_{i,k,l}$: polarization vector associated with the l-th propagation path.
- $\mu_{eg}^T \epsilon_{i,k,l}$: projection of the atomic transition dipole moment onto the polarization direction of the incident electromagnetic field.
- $\alpha_{l,k}$: complex gain of path l.
- $e^{-j(i-1)\phi_{l,k}}$: spatial phase progression across the array.

Thus the atomic channel is not written as a generic i.i.d. Rayleigh matrix. It explicitly includes atomic coupling, spatial propagation and multipath.

The important atomic coupling structure is

$$\boxed{\frac{\mu_{eg}^T \epsilon_{i,k,l}}{\hbar}}$$

which is a multiplicative coupling factor.

4.2 Equation (8): raw nonlinear measurement

For pilot $s_{k,p}$ from user k at time p,

$$y_{i,p} = \left| \sum_{k=1}^{K} g_{i,k} s_{k,p} + b_{i,p} + n_{i,p} \right|. \tag{8}$$

This is the fundamental difficulty of the problem.

Define

$$a_{i,p} = \sum_{k=1}^{K} g_{i,k} s_{k,p}. \tag{4}$$

Then

$$y_{i,p} = |a_{i,p} + b_{i,p} + n_{i,p}|.$$

Unlike a conventional coherent receiver, the atomic receiver does not directly output the complex quantity $a_{i,p}$. Instead, it outputs its magnitude after combination with the reference and noise.

4.3 Equation (9): reference signal

The known reference signal received at vapor cell i is

$$b_{i,p} = \frac{s_{b,p}}{\hbar} \mu_{eg}^T \epsilon_{b,i} \alpha_b e^{-j(i-1)\phi_b}. \tag{9}$$

Here:

- $s_{b,p}$ is the known reference signal at pilot time p.
- $\hbar$ is the reduced Planck constant.
- μ_{eg} is the atomic transition dipole moment.
- $\epsilon_{b,i}$ is the polarization vector of the reference field.
- α_b is the reference path gain.
- ϕ_b is the spatial phase shift of the reference signal.

The atomic coupling part is therefore

$$\boxed{\frac{\mu_{eg}^T \epsilon_{b,i}}{\hbar}.}$$

The reference signal is deliberately made strong and known so that it can assist the recovery of information hidden by the magnitude operation.

5 Strong-Reference Approximation

The key assumption is

$$|b_{i,p}| \gg |a_{i,p} + n_{i,p}|. \tag{5}$$

Starting from Eq. (8),

$$y_{i,p} = |a_{i,p} + b_{i,p} + n_{i,p}|. \tag{6}$$

Factor out the reference amplitude:

$$y_{i,p} = |b_{i,p}| \left| 1 + \frac{a_{i,p} + n_{i,p}}{b_{i,p}} \right|. \tag{7}$$

Using the magnitude expansion under the strong-reference condition,

$$\sqrt{1+x} \approx 1 + \frac{x}{2}, \tag{8}$$

the nonlinear magnitude measurement can be approximated by a linear real-valued model.

5.1 Equation (10): Taylor approximation

The measurement becomes approximately

$$y_{i,p} \approx |b_{i,p}| + \mathrm{Re}\left(\frac{a_{i,p} + n_{i,p}}{e^{jz_{i,p}}}\right), \tag{9}$$

where

$$z_{i,p} = \angle b_{i,p}. \tag{10}$$

The reference phase effectively rotates the observation.

5.2 Equation (11): linearized scalar model

The paper writes the resulting model as

$$y_{i,p} = \mathrm{Re}\left(e^{-jz_{i,p}} \sum_{k=1}^{K} g_{i,k} s_{k,p}\right) + |b_{i,p}| + \bar{n}_{i,p}, \tag{11}$$

where

$$\bar{n}_{i,p} = \mathrm{Re}\left(e^{-j\angle b_{i,p}} n_{i,p}\right). \tag{11}$$

The effective noise follows a real Gaussian distribution with variance $\sigma^2/2$.

Physical meaning. The strong reference acts like a phase reference. It converts the original nonlinear magnitude observation into an approximately linear measurement of a particular real projection of the complex received signal.

6 Matrix Measurement Model

Arrange all measurements into matrices.

Define

$$G \in \mathbb{C}^{I\times K}, \qquad S \in \mathbb{C}^{K\times P}.$$

The (i,k) element of G is $g_{i,k}$ and the (k,p) element of S is $s_{k,p}$.

6.1 Equation (12): matrix form

The complete linearized model is

$$Y = \mathrm{Re}\left[Z \circ (GS)\right] + |B| + N, \tag{12}$$

where $\circ$ denotes the Hadamard product.

The phase matrix is

$$[Z]_{i,p} = e^{-jz_{i,p}}. \tag{12}$$

Thus the original nonlinear problem

$$Y = |GS + B + N|$$

has been approximately transformed into

$$\boxed{Y = \mathrm{Re}[Z \circ (GS)] + |B| + N.}$$

This transformation makes gradient-based optimization possible.

7 Gradient Descent for the 1D Channel

7.1 Equation (13): least-squares objective

The channel estimation problem becomes

$$\hat{G} = \arg\min_{G} \mathcal{L}(G), \tag{13}$$

where

$$\mathcal{L}(G) = \|Y - |B| - \mathrm{Re}\,[Z \circ (GS)]\|_F^2 . \tag{13}$$

This asks:

> Find the complex channel G whose predicted atomic measurements best match the observed measurements.

7.2 Equation (14): gradient descent update

$$G^{(t+1)} = G^{(t)} - \eta_t \nabla_{G^*} \mathcal{L}(G^{(t)}), \tag{14}$$

where η_t is the step size.
The basic idea is

$$\boxed{\text{Current channel estimate} \rightarrow \text{prediction error} \rightarrow \text{gradient} \rightarrow \text{improved estimate.}}$$

7.3 Equation (15): gradient

The gradient used in the paper is

$$\nabla_G \mathcal{L}(G^{(t)}) = -2 \left(\left[Y - |B| - \mathrm{Re} \left((G^{(t)} S) \circ Z \right) \right] \circ Z^* \right) S^H. \tag{15}$$

Interpretation. Each iteration has the standard optimization structure:

$$\boxed{\text{prediction} \rightarrow \text{residual} \rightarrow \text{gradient} \rightarrow \text{channel correction.}}$$

8 Extension to a 2D Atomic Array

A 2D array has $I_1 \times I_2$ vapor-cell elements.

8.1 Equation (16): 2D multipath channel

The channel between user k and vapor cell (i_1, i_2) is

$$g_{i_1,i_2,k} = \sum_{l=1}^{L_k} \frac{1}{\hbar} \mu_{eg}^T \epsilon_{i_1,i_2,k,l} \alpha_{l,k} e^{-j[(i_1-1)u_{l,k} + (i_2-1)v_{l,k}]}. \tag{16}$$

The spatial frequencies are

$$u_{l,k} = \frac{2\pi d_1}{\lambda} \cos\theta_{l,k}, \tag{14}$$

and

$$v_{l,k} = \frac{2\pi d_2}{\lambda} \sin\theta_{l,k} \cos\phi_{l,k}. \tag{15}$$

Here:

- d_1, d_2: array spacings,
- λ: wavelength,
- $\theta_{l,k}$: elevation angle,
- $\phi_{l,k}$: azimuth angle.

Again, the atomic coupling factor is

$$\boxed{\frac{\mu_{eg}^T \epsilon_{i_1,i_2,k,l}}{\hbar}.}$$

8.2 Why low rank appears naturally

For a fixed user k, each propagation path contributes a separable spatial pattern. The user-specific channel slice can therefore be expressed as a sum of L_k structured components.

Hence

$$\operatorname{rank}(G_k) \le L_k. \tag{16}$$

This is the main reason PGD is useful in 2D.

8.3 Equation (17): 2D reference

The reference signal received at vapor cell (i_1, i_2) is

$$b_{i_1,i_2,p} = \frac{s_{b,p}}{\hbar} \mu_{eg}^T \epsilon_{b,i_1,i_2} \alpha_b e^{-j[(i_1-1)u_b+(i_2-1)v_b]}. \tag{17}$$

The corresponding atomic coupling term is

$$\boxed{\frac{\mu_{eg}^T \epsilon_{b,i_1,i_2}}{\hbar}.}$$

8.4 Equation (18): 2D linearized scalar measurement

$$y_{i_1,i_2,p} = \operatorname{Re}\left(e^{-jz_{i_1,i_2,p}} \sum_{k=1}^{K} g_{i_1,i_2,k} s_{k,p}\right) + |b_{i_1,i_2,p}| + n_{i_1,i_2,p}. \tag{18}$$

9 Tensor Representation

Define

$$\mathcal{Y}, \mathcal{Z}, \mathcal{G}, \mathcal{B}, \mathcal{N}$$

as compatible tensors.

The 2D channel tensor is

$$\mathcal{G} \in \mathbb{C}^{I_1 \times I_2 \times K}.$$

9.1 Equation (19): tensor model

The tensor measurement model can be written as

$$\mathcal{Y} = \mathrm{Re}\left[\mathcal{Z} \circ (\mathcal{G} \times_3 S^T)\right] + |\mathcal{B}| + \mathcal{N}. \tag{19}$$

Here $\times_3$ denotes multiplication along the third tensor mode.

10 Tensor Unfolding

To make optimization easier, the tensor is unfolded along its third mode.

The mode-3 unfolded channel is

$$G_{(3)} \in \mathbb{C}^{K \times I_1 I_2}.$$

Similarly,

$$Y_{(3)}, Z_{(3)}, B_{(3)}$$

represent unfolded versions of the measurement-related tensors.

10.1 Equation (20): unfolded measurement model

The unfolded model becomes

$$Y_{(3)} = \mathrm{Re}\left[Z_{(3)} \circ (S^T G_{(3)})\right] + |B_{(3)}| + N_{(3)}. \tag{20}$$

This has essentially the same structure as the 1D model, but now the unknown channel contains the spatial information of a 2D array.

11 Low-Rank Constraint

For each user k, the corresponding row of $G_{(3)}$ can be reshaped into

$$G_k \in \mathbb{C}^{I_1 \times I_2}.$$

Because the physical channel is composed of only L_k propagation paths,

$$\mathrm{rank}(G_k) \le L_k. \tag{21}$$

This constraint is the difference between ordinary GD and PGD.

11.1 Equation (22): objective definition

Define

$$f(G_{(3)}) = \left\| Y_{(3)} - |B_{(3)}| - \mathrm{Re}\left[Z_{(3)} \circ (S^T G_{(3)})\right] \right\|_F^2. \tag{22}$$

12 Projected Gradient Descent (PGD)

PGD has two operations:

$$\boxed{\text{gradient step} \quad + \quad \text{projection onto physically motivated low-rank set.}}$$

12.1 Equation (23): PGD update

$$G_{(3)}^{(t+1)} = \mathrm{proj}_{\mathcal{X}}\left(G_{(3)}^{(t)} - \zeta_t \nabla_{G_{(3)}} f(G_{(3)}^{(t)})\right). \tag{23}$$

The feasible set is

$$\mathcal{X} = \left\{X \in \mathbb{C}^{K \times I_1 I_2} : \mathrm{rank}(\mathrm{mat}([X]_{k,:})) \leq L_k,\ \forall k\right\}. \tag{17}$$

12.2 Equation (24): gradient

$$\nabla_{G_{(3)}} f(G_{(3)}^{(t)}) = -S^* \left(\left[Y_{(3)} - |B_{(3)}| - \mathrm{Re}\left(Z_{(3)} \circ (S^T G_{(3)}^{(t)})\right)\right] \circ Z^*\right). \tag{24}$$

12.3 Equation (25): projection problem

For a tentative matrix M,

$$\mathrm{proj}_{\mathcal{X}}(M) = \arg\min_X \|X - M\|_F^2 \quad \text{s.t.} \quad X \in \mathcal{X}. \tag{25}$$

Thus PGD asks:

> After taking a gradient step, what is the closest channel estimate that still obeys the physical low-rank propagation structure?

12.4 Equation (26): SVD

For user k, reshape the corresponding row and compute

$$U_k \Sigma_k V_k^T = \mathrm{svd}\left(\mathrm{mat}([M]_{k,:})\right). \tag{26}$$

12.5 Equation (27): truncated-SVD projection

Keep only the largest L_k singular values:

$$[X]_{k,:} = \mathrm{vec}\left(\tilde{U}_k \tilde{\Sigma}_k \tilde{V}_k^T\right)^T, \tag{27}$$

where

$$\tilde{U}_k = [U_k]_{:,1:L_k},$$

$$\tilde{V}_k = [V_k]_{:,1:L_k},$$

and

$$\tilde{\Sigma}_k = [\Sigma_k]_{1:L_k,1:L_k}.$$

Simple intuition. SVD decomposes a matrix into spatial modes ordered from strongest to weakest. Truncated SVD keeps only the number of modes physically justified by the number of propagation paths.

Algorithm 1 Projected Gradient Descent for 2D Rydberg Atomic Channel Estimation

Require: $Y_{(3)}, Z_{(3)}, B_{(3)}, S$, ranks $\{L_k\}$, step sizes $\{\zeta_t\}$

Initialize $G_{(3)}^{(0)}$
for $t = 0, 1, \ldots, T-1$ **do**
  Compute gradient using (24)
  $M \leftarrow G_{(3)}^{(t)} - \zeta_t \nabla f(G_{(3)}^{(t)})$
  for $k = 1, \ldots, K$ **do**
    Reshape $[M]_{k,:}$ into an $I_1 \times I_2$ matrix
    Compute its SVD
    Keep the largest L_k singular values
    Vectorize the truncated matrix and store as $[X]_{k,:}$
  end for
  $G_{(3)}^{(t+1)} \leftarrow X$
  Stop if $\|G_{(3)}^{(t+1)} - G_{(3)}^{(t)}\|_F^2$ is below threshold
end for

13 Algorithm 1: PGD in Plain Mathematical Form

13.1 Complexity reported in the paper

For the 2D case:

$$\mathcal{O}\left(T\left[K \min(I_1 I_2^2, I_1^2 I_2) + PKI_1 I_2\right]\right). \tag{18}$$

For a 1D array, projection is unnecessary and complexity reduces to

$$\mathcal{O}(TPKI). \tag{19}$$

14 Why GS Works for 1D but PGD is Needed for 2D

14.1 Biased GS measurement model

The earlier *Towards Atomic MIMO Receivers* paper uses the phase-retrieval model

$$z = |A^H s + b + w|. \tag{20}$$

For channel estimation, known pilots replace unknown symbols:

$$z_n = |S^H a_n + b_n + w_n|. \tag{21}$$

This is mathematically a phase-retrieval problem.

14.2 Core idea of GS

If the missing phase θ were known, then

$$z \circ e^{j\theta} \approx A^H s + b \tag{22}$$

would become linear.

GS alternates:

$$\boxed{\text{estimate phase} \rightarrow \text{solve linear least squares} \rightarrow \text{estimate phase again.}}$$

For biased GS, the update in the cited paper has the form

$$\theta^{(t)} = \angle(A^H s^{(t-1)} + b), \tag{23}$$

followed by

$$s^{(t)} = (AA^H)^{-1} A \left(z \circ e^{j\theta^{(t)}} - b \right). \tag{24}$$

14.3 GS initialization used in the cited paper

The cited paper constructs an augmented observation matrix and uses a spectral initialization:

$$M = \sum_{n=1}^{N} z_n \bar{a}_n \bar{a}_n^H. \tag{25}$$

Let v be its principal eigenvector. The scale is

$$\bar{r} = \frac{|v^H \bar{A}| z}{\|\bar{A}^H v\|_2^2}, \qquad \bar{s}^{(0)} = \bar{r} v. \tag{26}$$

The known bias entry is phase-normalized:

$$s^{(0)} = e^{-j\angle \bar{s}^{(0)}_{K+1}} \bar{s}^{(0)}_{1:K}. \tag{27}$$

14.4 GS versus GD versus PGD

Method	Main principle	Advantage	Limitation
GS	Alternates between phase recovery and linear estimation	Directly handles magnitude-only phase retrieval	Not naturally extendable to the 2D low-rank tensor structure
GD	Directly minimizes the linearized measurement residual	Simple and applicable to optimization formulation	Does not explicitly exploit low-rank spatial structure
PGD	GD followed by low-rank projection	Uses both measurement information and physical multipath structure	Requires SVD projection and rank knowledge/assumption

Important result reported by the paper. For 1D and many pilots, GD and GS become similar and approach CRLB. For small pilot length, GD outperforms GS in the reported simulations. For 2D, PGD can outperform plain GD because projection exploits the low-rank channel structure.

15 Worked GS Example

Consider a deliberately small illustrative problem:

$$A^H = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Suppose the unknown vector is

$$s = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Then

$$A^H s + b = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \qquad z = |A^H s + b| = \begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$

Assume an iterate

$$s^{(0)} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}.$$

The predicted complex field is

$$A^H s^{(0)} + b = \begin{bmatrix} 3.0 \\ 2.6 \end{bmatrix}.$$

Therefore the estimated phase is

$$\theta^{(1)} = \angle \begin{bmatrix} 3.0 \\ 2.6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Replace the predicted magnitude by the measured magnitude:

$$z \circ e^{j\theta^{(1)}} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$

Then solve the linear problem

$$s^{(1)} = (AA^H)^{-1} A(z - b).$$

This illustrates the GS logic: estimate phase from the current prediction, force the measurement magnitude, then return to the linear signal/channel domain.

16 Worked PGD Example

Suppose a user-specific 2×2 channel should have rank one:

$$G_{\text{true}} = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}, \qquad \text{rank}(G_{\text{true}}) = 1.$$

After one gradient step, assume we obtain

$$M = \begin{bmatrix} 2.1 & 1.9 \\ 0.9 & 1.2 \end{bmatrix}.$$

Because noise/gradient error has produced a full-rank matrix, PGD projects M onto the rank-one set:

$$M = U \Sigma V^T.$$

If

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}, \qquad \sigma_1 \gg \sigma_2,$$

then the rank-one projection is

$$\Pi_{\mathrm{rank}\le 1}(M) = \sigma_1 u_1 v_1^T.$$

Thus:

$$\boxed{\text{GD says: keep } M. \qquad \text{PGD says: keep only the physically meaningful dominant spatial mode.}}$$

17 CRLB Analysis

The Cramér–Rao Lower Bound provides a theoretical lower bound on the estimation error of unbiased estimators.

For the parameter vector $\boldsymbol{\theta}$, the Fisher information matrix is

$$\mathbf{J}(\boldsymbol{\theta}) = \mathbb{E}\left[\left(\frac{\partial \ln p(Y;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\left(\frac{\partial \ln p(Y;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)^T\right]. \tag{28}$$

The CRLB states

$$\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{J}^{-1}(\boldsymbol{\theta}). \tag{29}$$

For the unfolded channel-estimation model, the paper gives the CRLB benchmark based on the Fisher information associated with the real-valued linearized measurement model.

Conceptually,

$$\boxed{\text{More pilots} \Rightarrow \text{more Fisher information} \Rightarrow \text{smaller CRLB.}}$$

CRLB is not an algorithm. It is a benchmark: an unbiased estimator cannot achieve an error covariance below this bound.

18 MSE and NMSE

18.1 Mean-Squared Error

For one channel realization:

$$\mathrm{MSE} = \frac{1}{N_{\mathrm{par}}}\|\hat{G} - G\|_F^2. \tag{30}$$

For Monte-Carlo averaging:

$$\mathrm{MSE} = \mathbb{E}\left[\frac{1}{N_{\mathrm{par}}}\|\hat{G} - G\|_F^2\right]. \tag{31}$$

18.2 Normalized MSE used in the paper

The paper uses

$$\boxed{\mathrm{NMSE} = \frac{\mathbb{E}\{\|G_{(3)} - \hat{G}_{(3)}\|_F^2\}}{\mathbb{E}\{\|G_{(3)}\|_F^2\}}.} \tag{32}$$

NMSE in dB is

$$\mathrm{NMSE_{dB}} = 10\log_{10}(\mathrm{NMSE}). \tag{33}$$

Difference. MSE depends on the absolute channel scale. NMSE divides by channel energy, making different scenarios easier to compare.

19 Simulation Setup Reported in the Paper

The simulation section specifies:

- Rydberg states: $52D_{5/2}$ and $53P_{3/2}$.
- Transition angular frequency approximately
$$2\pi \times 5 \text{ GHz}.$$
- Transition dipole moment:
$$\mu_{eg} = [0, 1785.9qa_0, 0]^T.$$
-
$$q = 1.602 \times 10^{-19} \text{ C}.$$
-
$$a_0 = 5.292 \times 10^{-11} \text{ m}.$$
- Polarization components randomized with
$$\epsilon_{i,k,l}, \epsilon_{b,i} \sim \mathcal{N}(0, 1/3).$$
- Channel path gains:
$$\alpha_{l,k} \sim \mathcal{CN}(0, 1).$$
- Reference path gain:
$$\alpha_b \sim \mathcal{CN}(0, 10).$$
- Number of users:
$$K = 3.$$
- Random path count:
$$L_k \sim \text{Uniform}(3, 7).$$
- Pilots:
$$s_{k,p} \sim \mathcal{CN}(0, 1).$$
- PGD initialization:
$$[G_{(3)}^{(0)}]_{k,i} \sim \mathcal{CN}(0, 0.1).$$

20 How to Approach Fig. 3

Figure purpose: NMSE versus SNR for a 1D Rydberg atomic array.

20.1 Reported configuration

- 1D array with $I = 8$ vapor-cell elements.
- Compare GD, GS, and CRLB.
- Pilot examples discussed: $P = 10$ and $P = 30$.

20.2 Simulation procedure

For each SNR:

1. Generate random channel paths and construct G using (7).
2. Generate pilot matrix S.
3. Generate known reference B.
4. Generate nonlinear observations using (8).
5. Apply the strong-reference approximation/model for GD.
6. Estimate channel using GD equations (14)–(15).
7. Independently estimate channel using biased GS baseline.
8. Compute NMSE using (32).
9. Average over many Monte-Carlo realizations.
10. Compute CRLB using the corresponding 1D Fisher information model.

20.3 Expected qualitative result

With small pilot length, GD is reported to outperform GS. With sufficiently many pilots, GD and GS become similar and approach CRLB.

21 How to Approach Fig. 4

Figure purpose: NMSE versus SNR for a 2D Rydberg atomic array.

21.1 Reported configuration

$$I_1 \times I_2 = 8 \times 8.$$

Compare:

- GD without rank projection,
- PGD with truncated-SVD projection,
- CRLB.

21.2 Simulation procedure

For each SNR:

1. Generate a multipath 2D channel using (16).
2. Build the tensor model (19).
3. Unfold along mode 3 using (20).
4. Run ordinary GD using the gradient.
5. Run PGD using:

$$\text{gradient step} \rightarrow \text{reshape} \rightarrow \text{SVD} \rightarrow \text{truncate to } L_k \rightarrow \text{vectorize.}$$

6. Compute NMSE for both estimators.
7. Repeat Monte-Carlo trials and average.
8. Plot against SNR together with CRLB.

21.3 Expected qualitative result

For short pilots such as $P = 10$, PGD is better because low-rank projection adds useful structural information. For larger P, GD and PGD become similar.

22 How to Approach Fig. 5

Figure purpose: NMSE versus pilot length P at fixed SNR.

The paper uses

$$\mathrm{SNR} = 5 \text{ dB}$$

for the 2D-array pilot-length study.

22.1 Simulation procedure

1. Fix SNR at 5 dB.
2. Choose a range of pilot lengths P.
3. For each P, generate a fresh pilot matrix.
4. Generate observations.
5. Estimate the channel using GD and PGD.
6. Compute Monte-Carlo averaged NMSE.
7. Compute the corresponding CRLB.
8. Plot NMSE versus P.

22.2 Expected trend

$$P \uparrow \implies \text{more information} \implies \mathrm{NMSE} \downarrow .$$

The paper reports that pilot lengths above roughly 20 allow performance close to CRLB in the simulated 2D setting.

23 Assumptions Used in the Paper

1. **Perfect synchronization and coordination:**

   Pilot transmission is assumed synchronized and coordinated.
2. **Known pilots:**

   The receiver knows S.
3. **Known reference:**

   The reference signal parameters and phases are treated as known.

4. **Strong reference:**

   The approximation requires

$$|b_{i,p}| \gg |a_{i,p} + n_{i,p}|.$$

5. **Gaussian noise:**

   The relevant noise terms are modeled using Gaussian distributions.

6. **Finite multipath:**

   Each user has a finite number L_k of paths.

7. **2D low-rank spatial structure:**

   The user-specific channel slice has rank no larger than L_k.

8. **Independent random pilots:**

   The simulation uses i.i.d. complex Gaussian pilots.

24 Limitations and Important Caveats

1. The strong-reference approximation is not exact. Its validity depends on the reference dominating the useful signal and noise.
2. GD can converge to a poor solution if the step size is badly selected.
3. PGD requires a low-rank assumption. Incorrect knowledge of the effective path count can affect performance.
4. GS addresses phase retrieval directly but is not naturally formulated for the 2D low-rank tensor structure considered in this paper.
5. CRLB is a lower benchmark under its modeling assumptions and does not guarantee that a practical estimator will attain it.
6. The simulations depend on Monte-Carlo averaging and therefore require enough realizations for stable curves.

25 Recommended Simulation Platform

For reproducing Figs. 3–5, MATLAB is particularly convenient because the paper relies heavily on:

- matrix operations,
- complex Gaussian random variables,
- Frobenius norms,
- SVD,
- tensor reshaping,
- Monte-Carlo averaging.

A practical MATLAB project structure is

```
channel_estimation/
|
|-- main_fig3_1d.m
|-- main_fig4_2d.m
|-- main_fig5_pilot_length.m
|
|-- generate_channel_1d.m
|-- generate_channel_2d.m
|-- generate_pilots.m
|-- generate_reference.m
|-- generate_measurements.m
|
|-- estimate_GD_1d.m
|-- estimate_GD_2d.m
|-- estimate_PGD_2d.m
|-- estimate_GS_baseline.m
|
|-- compute_nmse.m
|-- compute_crlb.m
|-- project_low_rank.m
```

Python is equally possible, especially with NumPy/SciPy, but MATLAB's matrix and SVD workflow makes this particular paper convenient to implement.

26 Master Concept Map

Atomic multipath channel

$\downarrow$

Magnitude-only observation

$y = |Gs + b + n|$

$\downarrow$

Strong reference

$\downarrow$

Taylor linearization

$\downarrow$

$Y = \mathrm{Re}[Z \circ (GS)] + |B| + N$

$\downarrow$

1D	2D
$\downarrow$	$\downarrow$
GD / GS	GD / PGD
	$\downarrow$
	Low-rank SVD projection

$\downarrow$

NMSE comparison with CRLB

27 Final Practical Understanding

The paper's contribution can be remembered in one sentence:

> **A Rydberg atomic array produces a nonlinear magnitude-type channel observation; a strong reference approximately converts it into an optimization problem, and the 2D array's multipath geometry introduces a**

low-rank structure that PGD exploits through repeated truncated-SVD projection.

The roles of the algorithms are:

- **GS:** solves the phase-retrieval aspect by alternating between phase estimation and linear least-squares recovery.
- **GD:** directly minimizes the linearized measurement residual.
- **PGD:** performs GD but forces every 2D user-channel slice back onto the physically motivated low-rank manifold.
- **CRLB:** provides the theoretical lower benchmark.
- **NMSE:** measures normalized channel reconstruction error.